Week 03: two kinds of work

Programming task

Session 2: three short Python tasks.
Predict, change and repair.
Session 3: separate application.

Research task

Share Research 2 as a warm-up before Session 1.
Keep the feedback.
No new research task this week.

Use this guide with the same Week notebook; do not create a second set of weekly tasks. Email: submit.progmis@gmail.com.

Programming practice

Lists, dictionaries, for loops and accumulation
Predict → run → change → repair → explain

Three core tasks: understand the Python

Read and trace

Read list items by index, fields by key. Trace counts 2 + 3.

Change and observe

Change the second count to 4. Then try an empty list.

Repair the total

The loop uses = instead of +=. Repair it and retest.

Open Week_03.ipynb → Session 2. Keep a prediction, result and explanation for each task.

Core 1–2: predict, then change one thing

```
records = [{"name": "A", "count": 2},
           {"name": "B", "count": 3}]
print(records[0]["name"])
total = 0
for row in records:
    total += row["count"]
    print(row["name"], total)
print("total", total)
```

1. Explain [0] versus ["name"]; trace totals 2 then 5.
2. Change B to 4. For [], use Core 2 without [0] access.
3. Rerun initialization for each independent trial.

Use the complete Core 1 and Core 2 starters in the notebook; explain each changed result.

Core 3: find and repair one mistake

```
records = [{"name": "A", "count": 2},
           {"name": "B", "count": 3}]
total = 0
for row in records:
    total = row["count"]  # Deliberate overwrite bug.
print(total)
# Repair the update; retest the original and empty lists.
```

1. Predict the wrong total before running.
2. Repair the update; do not hard-code 5.
3. Retest the two records and an empty list.

Edit the matching Core 3 cell. Keep the wrong result, your repair and a retest.

Done: explain the concept, then save

Step	What to do / keep
Predict	Explain [0] versus ["name"] and the running totals.
Change	Keep changed-count and empty-list predictions/results.
Repair	Show original total 5 and empty total 0; explain +=.
Optional	Add C = 1 and trace each visit.
Save	Keep code and three records in Week_03.ipynb.

After practice, explain index versus key and adding versus overwriting.

Project application

Continue in the same notebook after the Python practice.

Apply the idea: three different totals

Step	What to do / keep
Open	Project studio → Independent lab: multi-batch planning note.
Edit	Complete the bread/rice loop: lines, containers and portions.
Given list	2 lines / 5 containers / 16 planned portions.
Empty / split bread	Empty: 0/0/0. Split counts 1+2: 3 lines / 5 containers / 16.
Write / save	A running trace, one field explanation and a review finding.

Keep the original project data and checks; Python practice uses a smaller list.

Research this week: share, then stop

Before Session 1 · Warm-up

Share Research 2: function inputs / return, wrong rule and a useful test.

Keep the feedback

Record one useful question and what you would clarify in your own explanation.

After class

No new research task. Finish the Week 3 work and prepare for Week 4.

Keep your existing Research 2 file. Do not create a seventh or replacement assignment.